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Early prediction of within-field variability wheat productive potential using Sentinel2 satellite data.

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Assessing agricultural production in the context of climate change is a global concern. In the recent decades, variable rate technology (VRT) for agricultural machinery has made it possible to adjust fertiliser rates on-the-go, allowing the within-field crop management. In this context, in order to select the most effective management practices, it is essential to identify the driving factors that determine yield variability, mapping the spatial distribution of these driving factors and to determine the local yield variability potential.

Mapping the homogeneous within-field areas of yield potential is used to define management zones. Remote sensing data provide a practical means of delineating these zones. The crop biophysical variable, cumulative evapotranspiration (ET_{cum}), derived from NDVI time series and climate data, was analysed to evaluate its ability to estimate yield. In the semi-arid conditions of the Spanish Central Plateau, wheat ET_{cum} maps were correlated with yield maps by non-linear regression with an R^2 of 88%. ET_{cum} serves as an effective proxy for yield estimation and the statistical analysis to determine the level of homogeneity within the field, the driving factors that determine yield variability, and mapping the spatial distribution of these driving factors. Nevertheless, the observed saturation effect in the biophysical variable highlights limitations that require further analysis.

Additionally, during the wheat season, expected potential yields can fluctuate in response to actual weather conditions. Consequently, updating yield predictions early in the season is critical for informed irrigation and fertilisation management decisions. The ability of ET_{cum} to forecast yields at early phenological stages, such as flag leaf and flowering—key stages for yield formation—is examined. Finally, the stability of spatial variability patterns, compared to those derived from ET_{cum} at maturity, is analysed as an indicator of the spatial distribution of yield drivers.

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